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Anytime Algorithm for Parametric Faults Accommodation under Handling Quality Constraints

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Abstract: Linear quadratic fault tolerant control is addressed. Under process and/or actuator faults, both fault accommodation and system reconfiguration procedures need the controller to be redesigned, which amounts to solving the algebraic Riccati equation associated with the post-fault system model. To overcome the risks of system instability and/or control inadmissibility that might result from the fault detection and isolation delay and the computation time of the post-fault control law, two progressive accommodation strategies, namely Newton-Kleinman and Matrix-Sign-Function, are investigated. The longitudinal control of an aircraft under actuator and structural faults is used to illustrate the approaches with respect to the handling quality constraints. *Copyright © 2009 IFAC*

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1. INTRODUCTION

Fault Tolerant Control (FTC) refers to the mechanism by which a given system is able to exhibit desired properties, both in normal operation and in the presence of faults occurring in sensors, actuators, or in the plant as well [Blanke et al., 2006]; FTC mechanisms may be implemented in practice either via pre-computed control laws or on-line automatic redesign.

Properties of interest in the analysis of the nominal and post-fault systems range from stability to optimality [Kwakernaak and Sivan, 1972]; analyzing optimality can be done using the Linear Quadratic (LQ) control technique. Regarding the post-fault behavior, an automatic control system is said to be linear quadratic fault tolerant as it provides that a quadratic objective can still be satisfied in the presence of faults belonging to a given set.

Obviously, in both cases, respectively nominal and faulty, the optimal control law of a system is computed upon the solution of a matrix algebraic Riccati equation (ARE) [Lancaster and Rodman, 1995]. However, it was shown in another paper that the successive application of ARE solutions provided at each step by an iterative technique is recommended as it improves the overall closed-loop behavior (see [Ciubotaru and Staroswiecki, 2009]).

This paper extends the previous work on progressive fault accommodation [Staroswiecki et al., 2007], as it compares the solutions provided by the Newton-Kleinman approach to obtaining the post-fault control accommodated solution with those yielded by another iterative solver based on the Matrix-Sign-Function; also, an application to the longitudinal control of an aircraft is developed, taking into account the handling criteria constraints that define the flight quality.

It is organized as follows: Section 2 introduces the nominal and fault tolerant linear quadratic control problems. Further, Section 3 sketches the progressive accommodation scheme and presents the two iterative methods for fault accommodation, while Section 4 states the numerical results and shows the simulations. Section 5 concludes the paper.

2. NOMINAL AND FAULT TOLERANT LINEAR QUADRATIC CONTROL

2.1 Nominal Operation

Consider the continuous linear time-invariant (LTI) deterministic system whose nominal operation is modeled by

$$M_n : \dot{x}(t) = A_n x(t) + B_n u_n(t), \text{ with } x(0) = x_0, \quad (1)$$

where $A_n \in \mathbb{R}^{n\times n}$ and $B_n \in \mathbb{R}^{n\times m}$ are the nominal system and control matrices, $x(t) \in \mathbb{R}^n$ and $u(t) \in \mathbb{R}^m$ are the state and control vectors, and $x_0 \in \mathbb{R}^n$ is the initial condition.

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2.2 Optimality

Define the quadratic cost functional as

$$J(x_0, u(t)) = \int_0^\infty \left[x^T(t)Qx(t) + u^T(t)Ru(t)\right] dt\,, \quad (2)$$

where $R > 0$ and $Q = \bar{Q}^T\bar{Q} \geq 0$ are weighting symmetric matrices, and $(.)^T$ stands for transposition.

Under the assumptions of stabilizability and detectability of pairs (A_n, B_n) and $(A_n, \bar{Q})$ respectively, the nominal linear state-feedback control law

$$u_n(t) = -K_n x(t)\,, \text{ with } K_n = R^{-1}B_n^T X_n\,, \quad (3)$$

is optimal, where $X_n \in \mathbb{R}^{n\times n}$ is the unique symmetric semipositive-definite solution of the nominal continuous algebraic Riccati equation (CARE)

$$CARE_n \,:\, A_n^T X_n + X_n A_n - X_n B_n R^{-1} B_n^T X_n + Q = 0\,. \quad (4)$$

In the nominal situation, the closed-loop system

$$\hat{M}_n \,:\, \dot{x}(t) = \hat{A}_n x(t)\,, \text{ with } \hat{A}_n = A_n - B_n K_n\,, \quad (5)$$

is stable; provided the solution X_n to (4), the quadratic cost in (2) takes the minimum value

$$J_{M_n}(x_0, u_n) \triangleq J_n = x_0^T X_n x_0 \;(\equiv J^\star)\,. \quad (6)$$

2.3 Handling Quality Constraints

Obviously, $u_n(t)$ ensures the stability of $\hat{M}_n$ and the optimality of J_n but the closed-loop system may still be constrained to behave in a class of admissible models, i.e., $\bar{\Psi}(\hat{M}_n) \leq 0$, where $\bar{\Psi}$ is some set of matrix inequalities defining the admissible closed-loop behaviors.

Depending on the application, performance requirements may particularly be expressed function of the damping coefficient and natural frequency of the natural modes of the system behavior; that is the matrix inequalities defining the class of admissible closed-loop models $\bar{\Psi}(\hat{M}_n) \leq 0$ will effectively reflect in conditions imposed onto the natural modes dynamic parameters, i.e., $\bar{\Psi}(\hat{\zeta}_n^i, \hat{\omega}_n^i) \leq 0$, thus providing an appealing way to defining system admissibility, where $i \geq 2$ stands for the index of the natural mode.

For example, in aircraft applications, the longitudinal motion is composed of two natural behaviors, namely the short-period (SP) mode respectively the long-period (phugoid (PH)) mode. In this regard, the flight quality (satisfactory, acceptable, poor) is described by the handling criteria constraints delimiting the numerical values of the dynamic parameters (damping coefficient, natural frequency) of both natural modes of evolution.

2.4 Faulty Operation

Since parametric faults are considered, the system to be controlled is described by the pair $(A_f, B_f) \neq (A_n, B_n)$, assuming that an LTI model is still able to capture the system behavior and an online Fault Detection, Identification and Estimation (FDIE) module provides (A_f, B_f). Regarding this, there are two different means by which the fault can be recovered according to the Fault Tolerance (FT) strategy used, namely the accommodation and the reconfiguration approaches.

Whatever the selected fault recovery technique, the best strategy consists of applying the optimal control solution to the system

$$M_f \,:\, \dot{x}(t) = A_f x(t) + B_f u_f(t)\,, \text{ with } x_{t_f} = x(t_f) \triangleq x_f\,, \quad (7)$$

on the post-fault time interval, i.e., $t \in [t_f, \infty)$.

Finding the optimal control associated with the faulty system is a controller redesign problem; it therefore amounts to solving the continuous ARE associated with the new situation, i.e.,

$$CARE_f \,:\, A_f^T X_f + X_f A_f - X_f B_f R^{-1} B_f^T X_f + Q = 0\,. \quad (8)$$

Assuming that the pairs (A_f, B_f) and $(A_f, \bar{Q})$ are still stabilizable and detectable respectively, the accommodated post-fault control law calculates as

$$u_f(t) = -K_f x(t)\,, \text{ with } K_f = R^{-1}B_f^T X_f\,, \quad (9)$$

and, similarly to (5), the post-fault closed-loop system behavior is modeled by

$$\hat{M}_f \,:\, \dot{x}(t) = \hat{A}_f x(t)\,, \text{ with } \hat{A}_f = A_f - B_f K_f\,; \quad (10)$$

furthermore, the control law (9) leads the cost functional (2) to the value

$$J_{M_f}(x_f, u_f) \triangleq J_f = x_0^T X_n x_0 + x_f^T (X_f - X_n) x_f\,. \quad (11)$$

This far, (1)-(6) and (7)-(11) define the FTC problem in the LQ framework (LQ-FTC).

2.5 Fault Tolerant Operation

However, the fact that solution $u_f(t)$ achieves the objective J_f optimally does not mean that it is satisfactory. Indeed, the cost J_f might exceed some given upper limit $J_{\max} \geq J^\star$ or the behavior associated with the accommodated control $u_f(t)$ might be unacceptable ($\bar{\Psi}(\hat{M}_f) \not\leq 0$), thus denying system M_n to deserve the "fault tolerant" label w.r.t. the fault situation M_f.

Definition 1. (Admissibility). Assuming that a solution (u_n, u_f) to the LQ-FTC problem (1)-(11) exists, for all initial conditions x_0, it is admissible iff

$$J_f \leq \rho J^\star\,, \quad (12a)$$

$$(i).\ \bar{\Psi}\left(\hat{M}_n\right) \leq 0\,,\ (ii).\ \bar{\Psi}\left(\hat{M}_f\right) \leq 0\,, \quad (12b)$$

where (12a) defines the maximal performance degradation admitted in faulty situations, with $\rho = \frac{J_{\max}}{J^\star} \geq 1$, while (12b) indicates in cases (i) and (ii) whether the closed-loop system is admissible before respectively after the fault.

In this perspective, system (1) is fault tolerant w.r.t. the fault situation (7) iff either the accommodation or the reconfiguration problem has an admissible solution.

In what follows, two iterative techniques to solving the post-fault CARE, namely Newton-Kleinman (NK) and Matrix-Sign-Function (MSF), are detailed in order to construct the progressive scheme for fault accommodation.

3. NK VS. MSF PROGRESSIVE ACCOMMODATION

3.1 Matrix-Sign-Function Approach

Consider the Hamiltonian-matrix (13)

$$H_f = \begin{bmatrix} A_f & -G_f \\ -Q & -A_f^T \end{bmatrix}\,, \text{ with } G_f = B_f R^{-1} B_f^T\,. \quad (13)$$

Then, matrix X_f is the desired solution to the post-fault CARE iff

$$\begin{bmatrix} A_f & -G_f \\ -Q & -A_f^T \end{bmatrix} = \begin{bmatrix} I_n & 0_n \\ X_f & I_n \end{bmatrix} \begin{bmatrix} \hat{A}_f & -G_f \\ 0_n & -\hat{A}_f^T \end{bmatrix} \begin{bmatrix} I_n & 0_n \\ X_f & I_n \end{bmatrix}^{-1},$$

where $\hat{A}_f = A_f - B_f K_f = A_f - G_f X_f$ is the closed-loop matrix in the faulty case, with I_n and 0_n respectively the identity and null matrices.

Henceforth, introduce the matrix

$$J = \begin{bmatrix} 0_n & I_n \\ -I_n & 0_n \end{bmatrix} \in \mathbb{R}^{2n \times 2n}$$

and observe that H_f is J-symmetric, i.e., $JH_f = (JH_f)^T$; that is if $\lambda_f \in \Lambda(H_f)$ then $-\lambda_f \in \Lambda(H_f)$. Hence, the spectrum of the Hamiltonian-matrix is symmetric w.r.t. the imaginary-axis and therefore H_f has exactly n stable eigenvalues.

Remember also that there always exists a similarity transformation T_f of eigenvectors such that

$$H_f = T_f \begin{bmatrix} J_f^- & 0_n \\ 0_n & J_f^+ \end{bmatrix} T_f^{-1}, \tag{14}$$

where J_f^- and J_f^+ are Jordan-blocks associated with the stable and unstable eigenvalues respectively.

The sign-function matrix of H_f may then be computed upon (14) as follows (see, e.g., [Roberts, 1980])

$$\operatorname{sign}(H_f) = T_f \begin{bmatrix} -I_n & 0_n \\ 0_n & I_n \end{bmatrix} T_f^{-1}; \tag{15}$$

note that $\Lambda(\operatorname{sign}(H_f))$ contains only ± 1's, which means the sign-function matrix (15) is involutory, i.e., $(\operatorname{sign}(H_f))^2 = I_{2n}$.

Thus, the Newton iteration for computing the principal square-root of the identity matrix

$$Z_{f_{k+1}} = \frac{1}{2}\big(Z_{f_k} + Z_{f_k}^{-1}\big), \text{ for } k = 1, 2, \ldots, N, \tag{16}$$

with the initial step $Z_{f_0} = H_f$, obeys $\lim_{k\to\infty} Z_{f_k} = \operatorname{sign}(H_f) = Z_{f_N}$ (see, e.g., [Higham, 1997]).

Moreover, accelerating the speed of the first steps is possible through an appropriate scaling [Byers, 1987] and increasing efficiency is reached by symmetrizing the iterations [Bierman, 1984], that is replacing (16) with

$$\bar{Z}_{f_k} = Z_{f_k}\big(\det(Z_{f_k})\big)^{-\frac{1}{2n}}, \tag{17a}$$

$$Z_{f_{k+1}} = \bar{Z}_{f_k} - \frac{1}{2}\big(\bar{Z}_{f_k} - (J\bar{Z}_{f_k})^{-1}J\big). \tag{17b}$$

Partitioning Z_{f_k} accordingly, the solution X_{f_k} to (8) results then by solving the system

$$\begin{bmatrix} Z_{f_k}^{12} \\ Z_{f_k}^{22} + I_n \end{bmatrix} X_{f_k} = -\begin{bmatrix} Z_{f_k}^{11} + I_n \\ Z_{f_k}^{21} \end{bmatrix}, \tag{18}$$

for $k = N$ (see, e.g., [Kenney and Laub, 1995]).

3.2 Newton-Kleinman Approach

Let X_{f_k} be the unique semipositive-definite solution of the algebraic Lyapunov equation

$$\hat{A}_{f_k}^T X_{f_k} + X_{f_k}\hat{A}_{f_k} + K_{f_k}^T R K_{f_k} + Q = 0. \tag{19}$$

Then, $\lim_{k\to\infty} X_{f_k} = X_{f_N}$, with the ordering $0 \le X_{f_N} \le \ldots \le X_{f_{k+1}} \le X_{f_k} \le \ldots \le X_{f_1}$ (see [Kleinman, 1968]), where, for $k = 1, 2, \ldots, N$, recursively,

$$K_{f_k} = R^{-1}B_f^T X_{f_{k-1}},$$

with $\hat{A}_{f_k} = A_f - B_f K_{f_k}$ and K_{f_0} chosen such that the matrix $\hat{A}_{f_0} = A_f - B_f K_{f_0}$ has stable eigenvalues. (In general, $X_{f_0} \ge X_{f_1}$ is not true.)

It follows that, given any initial stabilizing matrix K_{f_0}, for N large enough, the solution to the linear algebraic Lyapunov equation

$$\hat{A}_{f_N}^T X_{f_N} + X_{f_N}\hat{A}_{f_N} + K_{f_N}^T R K_{f_N} + Q = 0 \tag{20}$$

converges to that of the algebraic Riccati equation

$$A_f^T X_{f_N} + X_{f_N} A_f - K_{f_N}^T R K_{f_N} + Q = 0; \tag{21}$$

removing index N, remark that (21) and (8) are identical.

3.3 Progressive Accommodation Scheme

As shown in another paper [Staroswiecki et al., 2007], the progressive approach for fault accommodation would avoid applying the nominal control law (3) during the period $t_{ftc} - t_{fdie} = \bar{\Delta}$ $\big(\bar{\Delta} = \sum_{k=1}^{N} \Delta_k$ and $\bar{\Delta}_k = \sum_{l=1}^{k} \Delta_l\big)$ until the accommodated feedback is available, where t_{fdie} is the time at which the new post-fault pair (A_f, B_f) is given by the FDIE module while t_{ftc} is the instant at which the new post-fault control matrix K_f computed by the FAR (Fault Accommodation/Reconfiguration) module becomes available.

On the contrary, it would apply the control laws $u_{f_k}(t) = -K_{f_k}x(t)$ for $k = 1, 2, \ldots, N$, with K_{f_k}'s computed upon the solutions X_{f_k} provided either by (18) or (19), rather than waiting for the complete convergence of iterations. (Assume that N is the number of steps needed to obtain a satisfactory approximation either for the solution to (17) or (20) and take Δ_k the duration of iteration k.)

Therefore, using the Progressive Accommodation Scheme results in a three-stage post-fault behavior, as sketched in Algorithm 1.

Algorithm 1. Progressive Accommodation Scheme (PAS)

(1-1) During the first iteration,
$$\dot{x}(t) = (A_f - B_f K_n)x(t), \text{ for } t \in [t_f, t_{fdie} + \Delta_1).$$

(1-2) While progressive accommodation proceeds,
$$\dot{x}(t) = (A_f - B_f K_{f_k})x(t), \text{ for } t \in [t_{fdie} + \bar{\Delta}_k, t_{fdie} + \bar{\Delta}_{k+1}).$$

(1-3) When iterations reached convergence,
$$\dot{x}(t) = (A_f - B_f K_{f_N})x(t), \text{ for } t \in [t_{fdie} + \bar{\Delta}, \infty).$$

3.4 Anytime Progressive Accommodation

However, in real-time operation there are cases when the computation time needed to obtain precise or optimal solutions usually reduce the overall utility of the control system, which might become critical in certain failure conditions. That conducts to the concept of anytime algorithms which are algorithms whose quality of results improves gradually as computation time increases.

As not every algorithm that produces a series of approximate answers is a well-behaved anytime algorithm, a set of desired properties include the following (see [Zilberstein, 1996]):

- *measurable quality*: the quality of an approximate result can be determined precisely;
- *recognizable quality*: the quality of an approximate result can easily be determined at run-time within a constant time;
- *monotonicity*: the quality of the result is a nondecreasing function of time and input quality;
- *consistency*: the quality of the results is correlated with the computation time and input quality;
- *diminishing results*: the improvement in solution quality is larger at the early stages of the computation and it diminishes over time;
- *interruptibility*: the algorithm can be stopped at any time and provide some answer;
- *preemptability*: the algorithm can be suspended and resumed with minimal overhead.

In this respect, for the Newton-Kleinman approach used in progressive fault accommodation, the quality metric which monitors the progress in problem solving and allocates computational resources effectively that would naturally be imposed is the accuracy, measuring the distance between the solutions to the intermediate algebraic Lyapunov equations and the solution to the original algebraic Riccati equation.

Naturally, the solutions obtained with this technique are inputs to the result that interests the most, namely the cost of the post-fault accommodation, which is a measurable quality and also recognizable at the end of every sample Δ_k, provided the evaluation of the quadratic integral criterion. Also, the cost decreases over time and thus the quality of the control solution is non-decreasing as it is cheaper or at least equal to the previous one at every step, that being guaranteed by the monotonicity of the solutions of the series of algebraic Lyapunov equations. Moreover, the quadratic convergence rate provided by the Newton method is with respect to the diminishing results property. Finally, the approach is preemptive as long as it achieves stability after the first step provided the initial stabilization procedure and resuming it can be done using the control gain matrix obtained during the last iteration.

However, for the Matrix-Sign-Function approach used in progressive fault accommodation, the two desirable properties concerning monotonicity and the diminishing results are not fulfilled, because MSF is sign-alternating and does not preserve the semipositive-definiteness of the solutions to the post-fault CARE, thus invalidating it as an anytime algorithm.

In the sequel, the design of a linear-quadratic controller for the longitudinal model of a civil aircraft is considered, with the aim of proving the advantages of using the progressive accommodation approach after a structural and an actuator fault (see also [Ciubotaru and Staroswiecki, 2006]).

4. HANDLING QUALITY ACCOMMODATION OF B747 AIRCRAFT LONGITUDINAL MODEL

4.1 Flight Condition

Consider a Class III large transport aircraft Boeing 747 in Class I cruise flight. Let the Category B flight phase be, i.e., gradual maneuvering without precision tracking.

Assume a straight and constant level flight at 40000 ft fixed altitude and 0.8 Mach; the corresponding steady-state speed is 774 fps. (The CoG position is $(0.25, 0, 0)$ m.a.c. and the aircraft mass is 19792 $slug$.)

4.2 Nominal System

Open-Loop Model The aircraft longitudinal dynamics is described by the state vector $x = [\, u \; w \; q \; \theta \,]^T$, where u (ft/sec) and w (ft/sec) represent the inertial velocities in the x- and z- directions of F_B reference frame, and q (rad/sec) and θ (rad) represent respectively the pitch rate and the pitch angle (F_B denotes body-axis reference). The control input δ_E (rad) is the elevator deflection.

The linearized model of the system is given by the nominal pair of system and control matrices (A_n, B_n) as (see [Etkin and Reid, 1996])

$$A_n = \begin{bmatrix} -0.0069 & 0.0139 & 0.0 & -32.2000 \\ -0.0904 & -0.3147 & 773.9766 & 0.0 \\ 0.0001 & -0.0010 & -0.4284 & 0.0 \\ 0.0 & 0.0 & 1.0 & 0.0 \end{bmatrix},$$

$$B_n = [\, -0.0002 \; -18.0610 \; -1.1577 \; 0.0 \,]^T .$$

Closed-Loop Admissible System As far as aircraft control is concerned, admissibility is linked with the handling qualities, referring to those characteristics that govern the ability and precision with which the pilot performs his tasks.

It can be checked that the short-period mode provides the "best" performance as long as the damping coefficient ζ_{sp} is about 0.6 and the frequency ω_{sp} approximates at 3 rad/sec; there are no specific values for the dynamic parameters of the phugoid mode but only the damping coefficient ζ_{ph} to remain positive (see, e.g., [Roskam, 2001]).

In this respect, define the matrix inequalities of admissible behaviors particularly for the second-order closed-loop models associated with the full longitudinal model, respectively $\hat{M}_{sp}$ for the short-period mode and $\hat{M}_{ph}$ for the phugoid mode, i.e., $\bar{\Psi} = [\, \Psi_1 \; \Psi_2 \; \Psi_3 \; \Psi_4 \,]^T$, precisely

$$\Psi_1(\hat{M}_{sp}) = \frac{(\zeta_{sp} - 0.6)^2}{0.04} + \frac{(\omega_{sp} - 3)^2}{0.64} - 1 \leq 0\,, \quad (22a)$$

$$\Psi_2(\hat{M}_{sp}) = \frac{(\zeta_{sp} - 0.75)^2}{0.20} + \frac{(\omega_{sp} - 3.45)^2}{2.25} - 1 \leq 0, \quad (22b)$$

$$\Psi_3(\hat{M}_{sp}) = \frac{(\zeta_{sp} - 0.85)^2}{0.45} + \frac{(\omega_{sp} - 3.75)^2}{4.00} - 1 \leq 0, \quad (22c)$$

$$\Psi_4(\hat{M}_{ph}) = -\zeta_{ph} \leq 0\,; \quad (22d)$$

hence, conditions (22a), (22b), (22c) define all short-period closed-loops whose dynamic performance are respectively "satisfactory" if all hold, "acceptable" if only conditions (22b) and (22c) hold, "poor" if only condition (22c) holds, or "unacceptable" if none among (22a), (22b), (22c) holds; note also that (22d) turns to be linear and states if the control law provides an acceptable phugoid mode behavior.

To design the full-state linear quadratic regulator, the penalty matrices for the controls and the states are chosen such that $\bar{\Psi}(\hat{M}_n) \leq 0$ holds, precisely $Q = 10^{-8} \times I_4$ and $R = 10^{-4}$.

The nominal control matrix

$$K_n = [\,0.0076\ -0.0047\ -2.9529\ -3.0015\,] \tag{23}$$

provides the closed-loop system $\hat{M}_n$, with $\hat{A}_n = A_n - B_n K_n$, which has the stable spectrum

$$\Lambda(\hat{A}_n) = \{-2.0578 \pm 2.2020\imath, -0.0687 \pm 0.0907\imath\}\ .$$

The nominal operating modes, namely the short-period mode $\lambda^n_{sp_{1,2}} = -2.0578 \pm 2.2020\imath$, characterized by the dynamic parameters $(\zeta^n_{sp}, \omega^n_{sp}) = (0.6828, 3.0138)$, respectively the long-period mode (phugoid) $\lambda^n_{ph_{1,2}} = -0.0687 \pm 0.0907\imath$, with $(\zeta^n_{ph}, \omega^n_{ph}) = (0.6064, 0.1134)$, are in the area of Level 1 handling qualities for the aircraft longitudinal motion (see again [Roskam, 2001]).

4.3 Faulty System

Consider an actuator fault defined by the loss-of-effectiveness with $\tau_f = 50\,\%$ in the elevator deflection; that can be modeled in the post-fault system by multiplying the control matrix with a factor $(1 - \tau_f) = 0.5$, i.e., $B_f = (1 - \tau_f)B_n$. Also, assume a structural fault modeled in the system matrix as $A_f = A_n + \Delta A_f \times I_4$, with $\Delta A_f = \sigma^{\max}_n \times 10^{-3}$, where $\sigma^{\max}_n$ is the maximum singular value of A_n.

4.4 Progressive Accommodation

Newton-Kleinman Approach Note that, for the faulty pair (A_f, B_f) introduced before, the nominal control matrix K_n given in (23) does not ensure stability.

The post-fault closed-loop system $\hat{M}_f$, with $\hat{A}_f = A_f - B_f K_n$, has got an unstable pair of poles in the spectrum

$$\Lambda(\hat{A}_f) = \{-0.4139 \pm 1.8885\imath, 0.7110 \pm 0.0903\imath\}\ ;$$

the dynamic parameters for the phugoid mode are not acceptable, since $\zeta^f_{ph} = -0.9920 < 0$; also, the short-period characteristics, i.e., $(\zeta^f_{sp}, \omega^f_{sp}) = (0.2141, 1.9333)$, are not satisfactory.

In this situation, use the Bass-Armstrong stabilizing procedure (see again [Ciubotaru and Staroswiecki, 2009]) and obtain the initial control matrix

$$K_{f_0} = [\,7.0007\ -0.3173\ -4.4520\ 203.9306\,]$$

that provides the stable closed-loop system $\hat{M}_{f_0}$, with $\hat{A}_{f_0} = A_f - B_f K_{f_0}$, with two pairs of complex-conjugate poles

$$\Lambda(\hat{A}_{f_0}) = \{-0.7740 \pm 3.3699\imath, -0.7740 \pm 0.6729\imath\}\ .$$

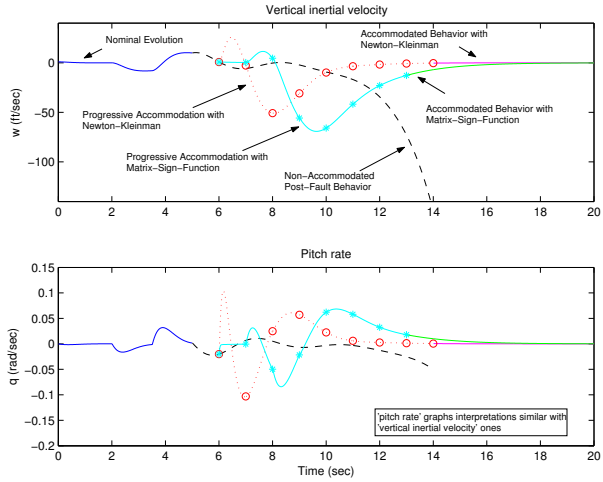

Fig. 1. B747-SP Progressive Fault Accommodation

To decide whether the progressive fault accommodation achieved convergence, a condition on the difference between the control gains in two successive iterations was used. The post-fault closed-loop needed $N = 8$ iterations for the accommodation.

Matrix-Sign-Function Approach Recall that there is no need for a special initialization procedure in this case. Using the same criterion for testing convergence as previously, the method took only $N = 6$ steps for the accommodation of the post-fault closed-loop system.

4.5 Accommodated System

Accommodated Closed-Loop The progressive fault accommodation, either based on the Newton-Kleinman approach or the Matrix-Sign-Function method, provides the control matrix

$$K^{acc}_f = [\,3.1203\ -0.0884\ -10.1884\ 33.0122\,]$$

which is computed based on the solution to the accommodated post-fault CARE; then, the accommodated closed-loop system $\hat{M}^{acc}_f$, with $\hat{A}^{acc}_f = A_f - B_f K^{acc}_f$, has a stable accommodated spectrum

$$\Lambda(\hat{A}^{acc}_f) = \{-1.4041 \pm 1.6043\imath, -0.7707 \pm -0.0607\imath\}\ .$$

Closed-Loop Simulation Fig. 1 displays the simulations of the closed-loop behavior during the post-fault period when applying the Newton-Kleinman respectively Matrix-Sign-Function progressive control accommodation schemes.

For being concise, the simulations reveal the accommodated behavior only of the inertial vertical velocity and that of the pitch rate respectively (short-period mode); the excitation is produced for 3 *sec* by a 5 *deg* magnitude doublet-pulse.

The fault occurred at $t_f = 4$ *sec*, the estimation of the faulty model was ready at $t_{fdie} = 5$ *sec*, therefore there is one sample of uncertain evolution, until $t_{fdie+\Delta} = 6$ *sec*, when the solution to the initial step problem will be readily available, this one being applied until $t_{fdie+2\Delta} = 7$ *sec* (for simplicity, the case of samples $\Delta_k \equiv \Delta = 1$ *sec* was considered).

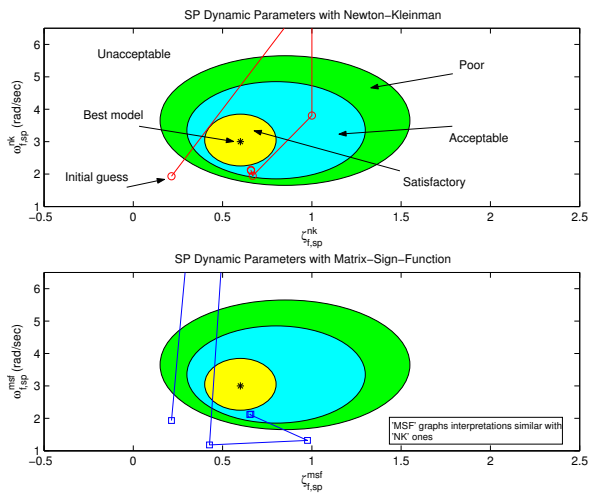

Fig. 2. B747-SP Dynamic Parameters Evolution

4.6 Fault Tolerance Analysis

Evaluate the Cost Functional For the initial condition $x_0 = [\,0.0215\ 0.9996\ -0.0001\ 0.0011\,]^T$, the minimum value of the quadratic cost is $J^\star = 3.6759 \times 10^{-9}$; at $t_f = 6$ *sec*, the post-fault "initial" condition estimates as $x_f = [\,0.1922\ 0.7962\ -0.0200\ 0.0044\,]^T$.

Using the solution to the accommodated post-fault CARE provides $J_f = 5.6881 \times 10^{-5}$ and thus $\rho = 1.5474 \times 10^4$; therefore, if the accommodated control is to be accepted then the maximum cost limit $J_{\max}$ should be fixed at a greater value than that of J_f.

Note also the large value of ρ which is associated with the FTC problem, showing the huge sensitivity of optimal control with respect to parameter variations.

Check the Closed-Loop Admissibility Fig. 2 shows the evolution of handling criteria for the short-period mode during the convergence of the iterative accommodation process.

Note that the Matrix-Sign-Function approach converges in less steps than the Newton-Kleinman method but see in Fig. 2 that MSF does several steps outside the "poor" performance ellipse while the NK iterates many times more inside the region of "acceptable" performance and close to the "satisfactory" performance ellipse.

However, against choosing Matrix-Sign-Function for progressive accommodation stands the fact that NK is an anytime algorithm, that is it improves the solution at each step, while no such result can be established for MSF.

5. CONCLUSION

Two iterative techniques to solving the continuous algebraic Riccati equation associated with the post-fault system in constrained operation, namely Newton-Kleinman (NK) and Matrix-Sign-Function (MSF), both used for progressive fault accommodation, are interrogated and analyzed in view of their application as anytime algorithms. The NK method is recommended since the value of the integral functional is non-increasing and available at each step, thus allowing the condition on the fault tolerance of the control solution to being evaluated with regard to the quadratic cost spent to compute it.

The MSF approach might appear simple for implementation (and somehow faster in number of steps performed, though the computational effort in the loop might not be comparable) but there is no mathematical rigor in applying it iteratively for anytime progressive accommodation (anyhow, even proceeding this way, the closed-loop progressive behavior is far from being acceptable).

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